# Appendix F. Classification problems Randomized algorithms Metrics

To understand the performance analysis of data structures like Treaps (chapter 3 – TODO: link) and Bloom filters (chapter 4 - TODO: link), we need to take a step back and first introduce a class of algorithms with which not every developer is familiar.

Most people, when prompted about algorithms, immediately think about deterministic algorithms. It's easy to mistake this subclass of algorithms for the whole universe: we expect an algorithm to be a sequence of instructions that, when provided a specific input, applies the same steps to return a well-defined output.

That's indeed the common scenario. It is possible, however, to describe an algorithm as a sequence of well-defined steps that will produce a deterministic result, but taking an unpredictable (though finite) amount of time. Even less intuitively, it is possible to describe algorithms that might produce a different, unpredictable result for the same input in every execution, or might also not terminate in a finite amount of time.

The first class of such algorithms is called *Las Vegas algorithms*, the second one is called *Monte Carlo algorithms*.